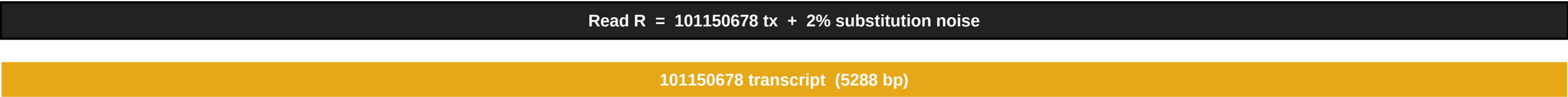


How a read gets assigned to a GOLGA8I copy – step by step

Real LOC101150678 transcript sampled with 2% noise, scored against all 7 copies' HMMs.

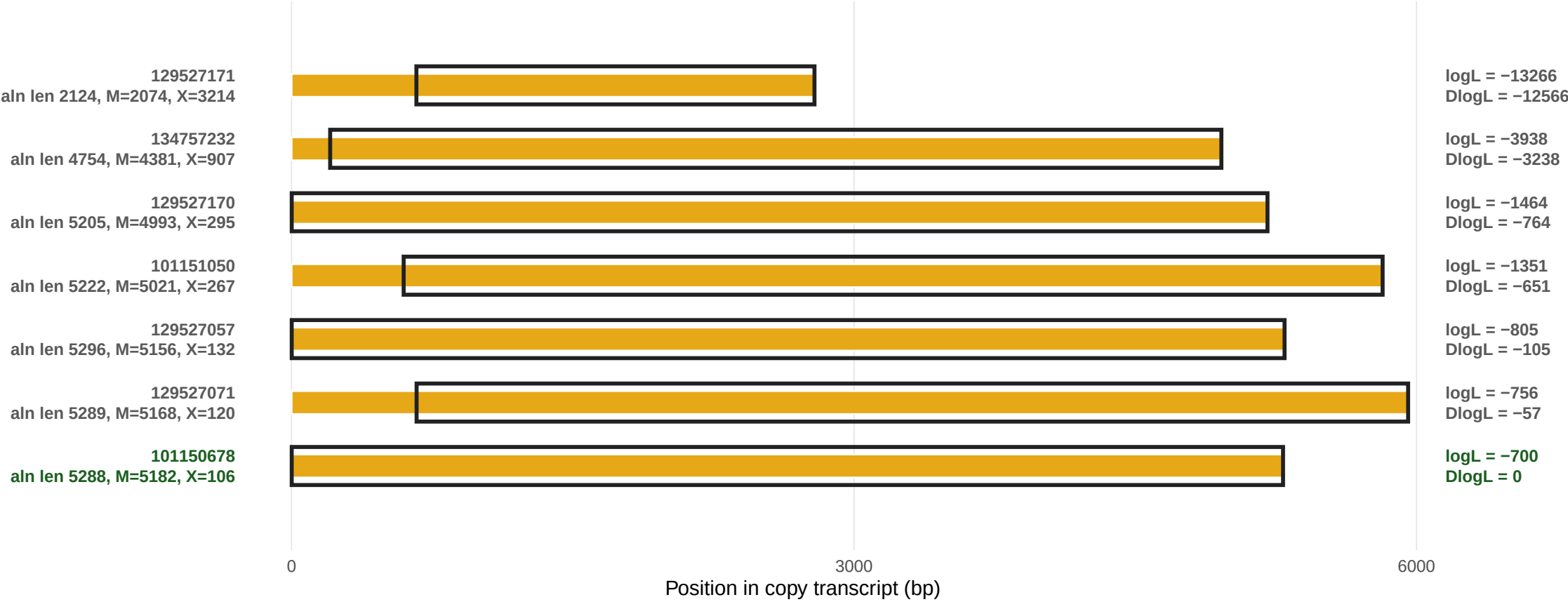
A The read

Full-length IsoSeq-style read sampled from one copy and perturbed at 2% positions. We will score this read against every copy's HMM.



B The 7 contestants – where R best aligns in every copy

Each row = one GOLGA8I copy. Yellow bar = full transcript. Black box = read R's best local alignment.
M = matches, X = mismatches (out of L = read length).



C How each number is computed

$$\log(\text{copy} \mid \text{read}) = \log\left(\frac{\text{p_match}}{\text{p_match} + \text{p_mismatch}}\right)$$

STEP 2 – Take logs:
log(0.95) = -0.0513
log(0.0167) = -4.0942

STEP 3 – Score per copy (L = read length):
logL(R | copy) = M . (-0.0513) + (L - M) . (-4.0942)

STEP 4 – Apply to GOLGA8I (L = 5288):
best : 101150678 M=5182 -> logL = -699.8
2nd : 129527071 M=5168 -> logL = -756.4 (DlogL = -56.6)
worst : 129527171 M=2074 -> logL = -13265.6 (DlogL = -12565.8)

D DlogL ranking

0 = best assignment, negatives = worse

